

Applying the Blind Spot Sieve Algorithm to Create Periodic Tables of Clustering Methods

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Abstract

The problem of identifying relevant scientific research directions is formalized using cluster analysis algorithms as an example by finding the intersections of knowledge domains defined by feature-based classifications. Using the study of knowledge domains in applied computer science as an example, intersections of feature spaces are constructed to build periodic tables. The "blind spots" of the resulting tables are used to identify relevant research topics and to propose concepts for their solutions. The blind spot sieve algorithm is based on the principle of intersecting classification features.

The fundamental significance of this work lies in the creation of a model for delineating domains of scientific inquiry defined by feature-based classifications through the application of a sieve algorithm that identifies relevant research based on a "blind spot". Its practical significance lies in the periodic tables of pixel clustering methods themselves, the identified relevant tasks, and the proposed concepts for their solutions.

Keywords: epistemological algorithm, blind spot sieve, periodic table, clustering methods, image segmentation, solution concepts, features.

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Introduction

The study of knowledge domains in applied computer science makes it possible to understand the organization of one of the forms of epistemological theory. Epistemological theory examines the possibilities of human cognition, the structure of cognitive activity, the forms and criteria of truth and reliability of knowledge, the nature of cognition and its limits, as well as methods and algorithms of cognition. An algorithm of cognition is an objective system of coherent cognitive actions independent of the will of the individual. The first aim of this study is to present the epistemological blind spot sieve algorithm for identifying relevant research topics at the intersection of cluster analysis and software engineering. The second aim is to apply the algorithm: based on the constructed periodic tables of clustering methods, existing solutions in the field of cluster analysis are systematized. The third aim is to create a visual model of data representation for understanding a scientific field in particular and for expanding worldview in general. The fourth aim is to introduce the principle of constructing periodic tables.

The first section examines the blind spot sieve algorithm, so named by analogy with the sieve of Eratosthenes, with the difference that the empty cells represent relevant research topics rather than the prime numbers sought. The steps of the blind spot sieve algorithm and its input and output data are described. The principle of constructing periodic tables, namely the intersection of features, is explained. The second section demonstrates the application of the blind spot sieve algorithm to identify relevant research topics at the intersection of the knowledge domains of cluster analysis, software engineering, and image types. An example is given of constructing a periodic table from systems of two, three, and four features for image pixel clustering methods. The third section uses the empty cells of the periodic tables ("blind spots") to reveal the identified relevant tasks and to propose concepts for their solutions based on considerations of symmetry, inversion, analogy, similarity, and intuition, including in relation to known solutions. The fourth section presents the conclusion, inference, and guidance of this study.

1. Blind Spot Sieve Algorithm

The blind spot sieve algorithm identifies all relevant research topics at the intersections of domains defined by feature-based classifications. It is based on a system of two principles: the intersection of classification features and their unfolding into a table. The algorithm for constructing the simplest periodic table for identifying "blind spots" consists of the following four steps:

1. An existing or a newly proposed adequate pair of principles for classifying objects of study into N and M feature (attribute) values is selected.

2. A single-level ($N \times M$) table is constructed from the intersection of attributes, in which the values of the classifying features are arranged horizontally and vertically, and cells are placed at the intersections for filling with objects of study having common properties derived from two different classifications with N and M features.
3. The classification ($N \times M$) table is filled with all known solutions.
4. Relevant research topics are identified from the cells that remain empty in the table ("blind spots").

As input, the blind spot sieve algorithm requires at least a classification pair and knowledge of known solutions satisfying the intersection of classification features. The output of the algorithm is a periodic table of known solutions containing a set of empty cells, or at least one empty cell. Once such a periodic table of solutions (objects of study) has been obtained, the question arises as to the expediency of conducting research to fill a specific "blind spot". If the periodic table is complete, this means that, for the given set of classification features, everything has already been explored. It is then necessary to select or devise another classification pair of features, that is, to change at least one of the two research domains, and repeat the algorithm.

2. Application of the Algorithm

Table 1 presents the simplest periodic table of clustering methods based on two feature classifications. Feature N , "software implementations", takes either the value "static" or "dynamic". Feature M , "cluster formation strategy", takes either the value "agglomerative", "iterative", or "divisive".

Table 1. Simple periodic ($N \times M$) table of clustering methods

	Static software implementations	Dynamic software implementations
Agglomerative cluster formation strategy	AACM (original, 2006 [1], modified, 2021 [2], 2024 [3])	Parameterized Ward's method (2015) [6]
Iterative cluster formation strategy		K-meanless (2013) [7] SI (2015) [8] in [6] CI (2018) [9] – improved [6] method
Divisive cluster formation strategy	LiMO (2024) [4, 5]	OptHstSegSu (2019) [10, 11] Otsu Hierarchical (2021) [12, 13]

According to the data represented by the "blind spot" in Table 1, the relevant line of research is the search for a software implementation of an iterative clustering algorithm with fixed allocated RAM for execution, in which the number of clusters in the partition remains constant throughout program execution. In effect, this refers to a program for improving the quality of a given cluster partition, with the specific feature that the number of steps required to reach the minimum approximation error value is known in advance. The creation of such a relevant software implementation is feasible due to the potential use of one-step optimization (reclassification) of groups of pixels forming individual clusters, as implemented in the divisive "static" LiMO method [4].

If the number of classification features is increased, the expanded periodic table reveals subtle differences between the desired and existing methods, thereby making it possible to develop concepts for new solutions.

Periodic Table 2, expanded to three features, includes the cluster analysis methods known to the author and the author's scientific supervisor that are applicable to the problem of image segmentation. In the "blind spot" cells for which no algorithm has been found, relevant tasks are indicated, and the concepts of their solutions are presented in Table 3.

With the introduction of the fourth classification feature, "image type", with the attribute values "color" and "grayscale", periodic Table 3 reveals more "blind spots" corresponding to relevant scientific research topics than Table 2. Although some methods can handle the task of clustering color pixels, the problem of "big data" becomes more significant in the development of relevant algorithms.

3. Relevant Tasks and Concepts for Their Solutions

Table 4 presents the relevant tasks identified from Table 2 and the concepts for their solutions. The left column of Table 4 lists the identified relevant tasks. The right column presents the proposed solution concepts.

Table 2. Single-level periodic table based on three features

	Hierarchical formation		Non-hierarchical formation	
	Static architecture	Dynamic architecture	Static architecture	Dynamic architecture
Agglomerative strategy	AACM [1, 2, 3] "3.3"	Ward's [4] Ward's param. method [5]	"3.4"	"3.7"
Iterative strategy	"3.5"	SI [6] CI [7]	"3.6"	K-meanless [8], K-means [9]
Divisive strategy	"3.2"	Otsu Hierarchical [10, 11]	LiMO [12, 13] "3.1", "3.2"	OptHstSegSu [14]

Table 3. Expanded periodic table of clustering methods based on four features

	Hierarchical formation				Non-hierarchical formation			
	Static architecture		Dynamic architecture		Static architecture		Dynamic architecture	
	Grayscale	Color	Grayscale	Color	Grayscale	Color	Grayscale	Color
Agglomerative strategy	AACM [1, 2, 3]	"3.3"	Ward's [4] Ward's param. method [5]	Ward's [4], Ward's param. method [5]	"3.4"	"3.7.1"	"3.7.2"	"3.7.2"
Iterative strategy	"3.5"		SI [6] CI [7]	SI [6] CI [7]	"3.6"	K-meanless [8] K-means [9]	K-meanless [8] K-means [9]	K-meanless [8] K-means [9]
Divisive strategy		"3.2"	Otsu Hierarchical [10, 11]		LiMO [12, 13] "3.1"	"3.2"	OptHstSegSu [14]	

4. Conclusion, inference and guidance

Conclusion: An increase in the number of classification features increases the depth of nesting of the periodic table and reveals new previously unknown aspects of earlier studied research topics. Using the increase in feature detail as an example, periodic Tables 2 and 3 were obtained from the simplest periodic Table 1. These tables revealed new relevant research topics, presented in the left column of Table 4. The right column of Table 4 presents the concepts proposed for their solutions.

Inference: The discovered blind spot sieve algorithm contributes to the theory and practice of cognition. It systematizes scientific inquiry through the delineation of domains and the predetermination of what is relevant (for example, see Table 3). As an epistemological algorithm, it makes it possible to identify relevant research topics at the intersections of knowledge domains defined by feature-based classification. The interdisciplinary significance of this finding is due the intersection of the disciplines of cluster analysis and computer engineering, as well as by the very nature of the work itself, namely the theory of cognition.

Guidance: The guidance of this discovery lies in the fact that all periodic tables are constructed according to the principle of intersecting classification features, with these features indicated along the horizontal and vertical layout of the resulting table. The most famous existing example that follows this principle is the periodic table of chemical elements developed by D.I. Mendeleev.

Table 4. List of identified relevant tasks and proposed solution concepts for promising solutions

Tasks ("3")	Solution concepts ("K")
"3.1": Investigation and testing of the architecture of a linear divisive multithreshold method for clustering grayscale images	"K.1": The solution concept [12, 13] consists in dividing clusters into nested ones down to the original ones by fixing previously identified thresholds, with the possibility of constructing a binary tree [15]. Replacing the functional will contribute to the development of a similar range of solutions for the linear multithreshold Otsu [16], Kapur [17], and Kittler [18] methods (multithreshold Otsu, Kapur and Kittler methods).
"3.2": Search for the architecture of a linear divisive multithreshold clustering method for color images, consisting in an improved extrapolation of the architecture of the "linear lineup" of methods to the case of a color image	"K.2": The solution concept can be seen in the search for suboptimal solutions through a sequence of forced dichotomous separations of several "working ranges" from "individual clusters" simultaneously, while skipping the standard procedures for finding the optimal threshold for partitioning into nested cluster pairs. This yields a histogram-based technology with forced acceleration of separations without calculating optima in the initial and intermediate computation cycles.
"3.3": Search for the architecture of agglomerative clustering of color images	"K.3.1": The first solution to this task can be seen in the search directions for a fast suboptimal method through the forced merging of all consecutive pairs of adjacent clusters at the initial and intermediate stages of cluster enlargement. In effect, this involves a modification of the Arifin–Asano agglomerative data clustering method [1, 2, 3]. This yields a histogram-based technology with forced merging.
	"K.3.2": The second solution can be seen in the application of superpixel technology, that is, operation in terms of dynamic trees with enlarged and preliminarily prepared datasets (enlarged pixels).
"3.4": Search for non-hierarchical agglomerative clustering	"K.4": The solution concept can be seen in supplementing the Arifin–Asano agglomerative clustering method (AACM) [1, 2, 3] with an intermediate one-step reclassification of "individual clusters" to a previously identified set of means, as in the LiMO method [12, 13].

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References

- [1] Arifin AZ, Asosno A. Image segmentation by histogram thresholding using hierarchical cluster analysis. *Pattern Recognition Letters* 2006; 27(13): 1515-1521. DOI: 10.1016/j.patrec.2006.02.022.
- [2] Khanykov IG. Technology modification for distance calculation between pairs of adjacent clusters in multilevel thresholding. *Trudy MAI* 2021; 118: 1-34. DOI: 10.34759/trd-2021-118-13.
- [3] Khanykov IG., Nenashev VA., Kharinov MV. Segmentation of Earth remote sensing images based on agglomerative pixel clustering using the minimum increment of the total squared error as a decision function. *Computer Optics* 2026; 50(1): 1-18. DOI: 10.18287/COJ1591.
- [4] Ward JHJ. Hierarchical grouping to optimize an objective function. *J. Am. Stat. Assoc.* 1963; 58(301): 236-244.
- [5] Khanykov IG, Kharinov MV, Patel C. Image Segmentation Improvement by Reversible Segment Merging. *Int. Conf. on Soft Computing and its Engineering Applications, icSoftComp-2017, IEEE Gujarat Section Proceedings, at CHARUSAT, Changa, Anand, India. 2017; 1-7. DOI: 10.1109/ICSOFTCOMP.2017.8280096.*
- [6] Kharinov MV, Khanykov IG. [Optimization of Piecewise Constant Approximation for Segmented Image] *Optimizaciya kusochno-postoyannogo priblizheniya segmentirovannogo izobrazheniya*. SPIRAS Proceedings 2015; 3(40): 183-202. DOI: 10.15622/sp.40.12. (In Russian).
- [7] Nenashev VA, Khanykov IG, Kharinov MV. A Model of Pixel and Superpixel Clustering for Object Detection. *J. Imaging* 2022; 8: 274. DOI: 10.3390/jimaging8100274.
- [8] Dvoenko S. Meanless k-means as k-meanless Clustering with the bi-partial Approach. *Pattern Recognition and Information Processing (PRIP'2014): Proceedings of the 12th International Conference (28-30 May 2014, Minsk, Belarus)*. 50-54.
- [9] Hartigan JA., Wong MA. Algorithm AS 136: A K-Means Clustering Algorithm. *Journal of the Royal Statistical Society. Series C (Applied Statistics)* JSTOR 1979; 28(1): 100-108. DOI: 10.2307/2346830.
- [10] Kharinov MV. Model of the quasi-optimal hierarchical segmentation of a color image. *Journal of Optical Technology*. 2015; 82(7): 425-429. DOI: 10.1364/JOT.82.000425.
- [11] User: Khar, Source: <<http://www.machinelearning.ru/wiki/index.php?title=user:Khar>> (accessed on 22.05.2025), Source: <https://disk.yandex.ru/d/xnVb9F4_M1hzUw> (accessed on 22.05.2025).
- [12] Khanykov IG. [Linear multithreshold Otsu method] *Lineynyj mul'tiporogovyy metod Otsu*. [Proceedings of XXV All-Russian Conference of Young Scientists on Mathematical Modeling and Information Technologies] *XXV Vserossiyskaya konferenciya molodyh uchyonyh po matematicheskomu modelirovaniyu i informacionnym tekhnologiyam*. Novosibirsk. 2024; 62. (In Russian).
- [13] Khanykov IG. [Linier multithreshold Otsu method] *Lineynyj mul'tiporogovyy metod Otsu*. Presentation video. Source: <https://vkvideo.ru/playlist/-131900260_1/video-131900260_456239039?linked=1>, presentation starts from 1:31:50.

- [14] Kharinov MV. Image segmentation method by merging and correction of sets of pixels // Pattern recognition and image analysis 23, 393-401 (2013).
- [15] User: Khar, Source: <<http://www.machinelearning.ru/wiki/index.php?title=user:Khar>> (accessed on 22.05.2025), Source: <<https://disk.yandex.ru/d/67KoREKNGECWRA>> (accessed on 22.05.2025)

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Применение алгоритма решета «белых пятен» для создания периодических таблиц методов кластеризации

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Аннотация

Формализуется задача актуального научного поиска на примере алгоритмов кластерного анализа путем нахождения пересечения областей знаний, задаваемых признаковыми классификациями. На примере исследования областей знаний прикладных компьютерных наук выполняется пересечение признаковых пространств для построения периодических таблиц. По «белым пятнам» получаемых таблиц определяются актуальные темы исследований, предлагаются концепты их решений. Основа алгоритма решета «белых пятен» – принцип пересечения классификационных признаков.

Фундаментальная значимость работы лежит в создании модели разграничения областей научного поиска, задаваемых признаковыми классификациями, через применение алгоритма решета с выявлением актуального исследования по «белому пятну». Практической значимостью служат сами периодические таблицы методов кластеризации пикселей, выявленные актуальные задачи и предлагаемые концепты их решений.

Ключевые слова: гносеологический алгоритм, решето «белых пятен», периодическая таблица, кластерные методы, сегментация изображений, концепты решений, признаки.

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Сведения об авторе

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